How-To Library

Informatica

Configuring pushdown optimization for Microsoft Azure Synapse SQL using ODBC Connector

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Abstract

This article explains how to configure pushdown optimization for a mapping that uses the ODBC connection type to read from or write to Microsoft Azure Synapse SQL. This article also explains the pushdown optimization functions, transformations, and operators that you can push down to Microsoft Azure Synapse SQL using the Microsoft Azure Synapse SQL ODBC connection type.

Supported Versions

- Informatica Cloud® Data Integration Microsoft Azure Synapse SQL Connector

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Overview

When you enable pushdown optimization for a task, the task pushes down the entire transformation logic as SQL queries to the underlying database for processing and the database runs the query. Since the entire transformation logic is encapsulated in an SQL query and there is no back-and-forth data movement between the database and the Informatica engine, the task is processed faster.

It is important to know what your primary use case is before you configure pushdown optimization. If your use case is to process data in the cloud for a mapping that reads from a Cloud Data Warehouse or Cloud Data Lake and writes to a Cloud Data Warehouse target such as Microsoft Azure Synapse SQL, you can use pushdown optimization to push the processing logic to the warehouse or data lake using the applicable native connector as the source and Microsoft Azure Synapse SQL as the target. The native connectors offer a wide range of capabilities that you can explore. For more information about configuring pushdown optimization and the supported functionalities using Microsoft Azure Synapse SQL Connector, see the help for Microsoft Azure Synapse SQL Connector.

However, if you are using your on-premises data warehouse or any ODBC-based targets, ODBC based pushdown using the ODBC Connector is the better choice. You do not need any separate license to use ODBC pushdown.

In this article, we'll configure pushdown optimization for a task that uses ODBC Connector to push transformation logic to process in the Microsoft Azure Synapse SQL source and target databases. The ODBC connection must use the Microsoft Azure Synapse SQL subtype in the connection. We'll show you how to optimize a Microsoft Azure Synapse SQL ODBC task to read from or write to Microsoft Azure Synapse SQL. You can configure source or full pushdown for the task.

Perform the following steps to optimize a Microsoft Azure Synapse SQL ODBC task:

1. Configure the Microsoft Azure Synapse SQL ODBC driver.

2. Create an ODBC connection with the ODBC subtype as Microsoft Azure Synapse SQL and configure the Microsoft Azure Synapse SQL database properties to which you want to connect.
3. Create a mapping that uses the configured ODBC connection and enable the task for pushdown optimization.

When you run the task enabled with pushdown optimization, the task converts the transformation logic to an SQL query. The task sends the query to the database, and the database uses the database resources to run the query. For information about the advanced session properties that you can use with pushdown optimization using a Microsoft Azure Synapse SQL ODBC connection, see the topic "Advanced Session Properties" in Tasks in the Data Integration help.

Prepare for pushdown optimization using a Microsoft Azure Synapse SQL ODBC connection

Before you can configure pushdown optimization using a Microsoft Azure Synapse SQL ODBC connection, you must install the Microsoft ODBC Driver version 13 or later for SQL Server for Windows. Install the Microsoft ODBC Driver for SQL Server on the machine where the Secure Agent is installed.

Download the Microsoft ODBC driver from the following link: https://www.microsoft.com/en-us/download/details.aspx?id=53339

You cannot configure a Microsoft Azure Synapse SQL ODBC connection on SUSE Linux Enterprise version 12.0.

When you use the Microsoft ODBC Driver 17, you cannot run mappings on Red Hat Enterprise Linux 8.

Configure the Microsoft Azure Synapse SQL ODBC driver on Windows

Before you establish an ODBC connection to connect to Microsoft Azure Synapse SQL on Windows, configure the ODBC driver.

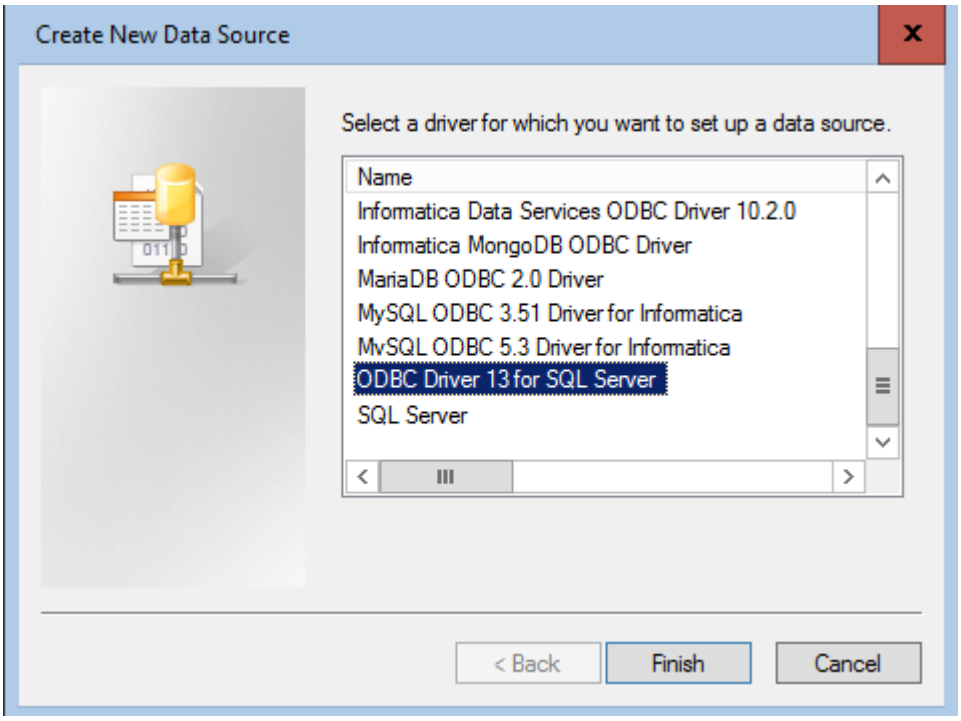
Perform the following steps to configure the ODBC driver:

1. Download the Microsoft Azure Synapse SQL ODBC 64-bit driver
2. Install the Microsoft Azure Synapse SQL ODBC driver on the machine where the Secure Agent is installed.
3. Open the folder in which ODBC data source file is installed.
4. Run the `odbcad32.exe` file under `C:\WINDOWS\System32`.

   The **ODBC Data Sources Administrator** box appears.
5. Click **System DSN**.

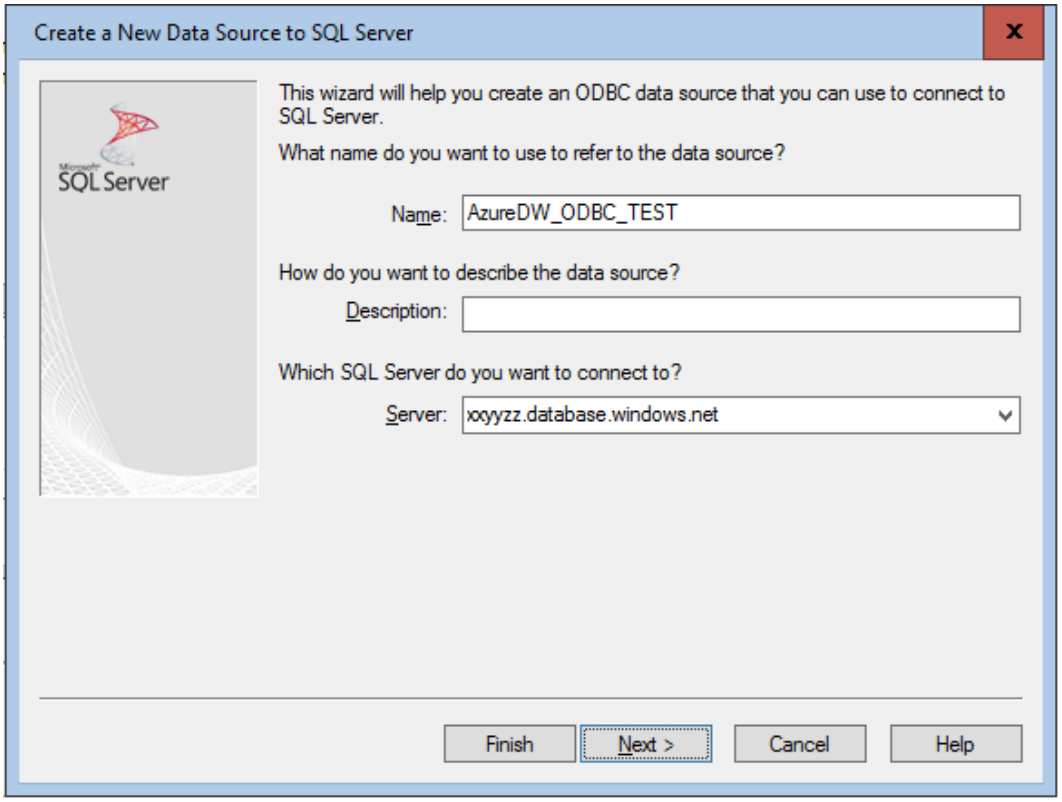
   The **System DSN** tab appears.
6. Click **Add**.

   The **Create New Data Source** box appears.



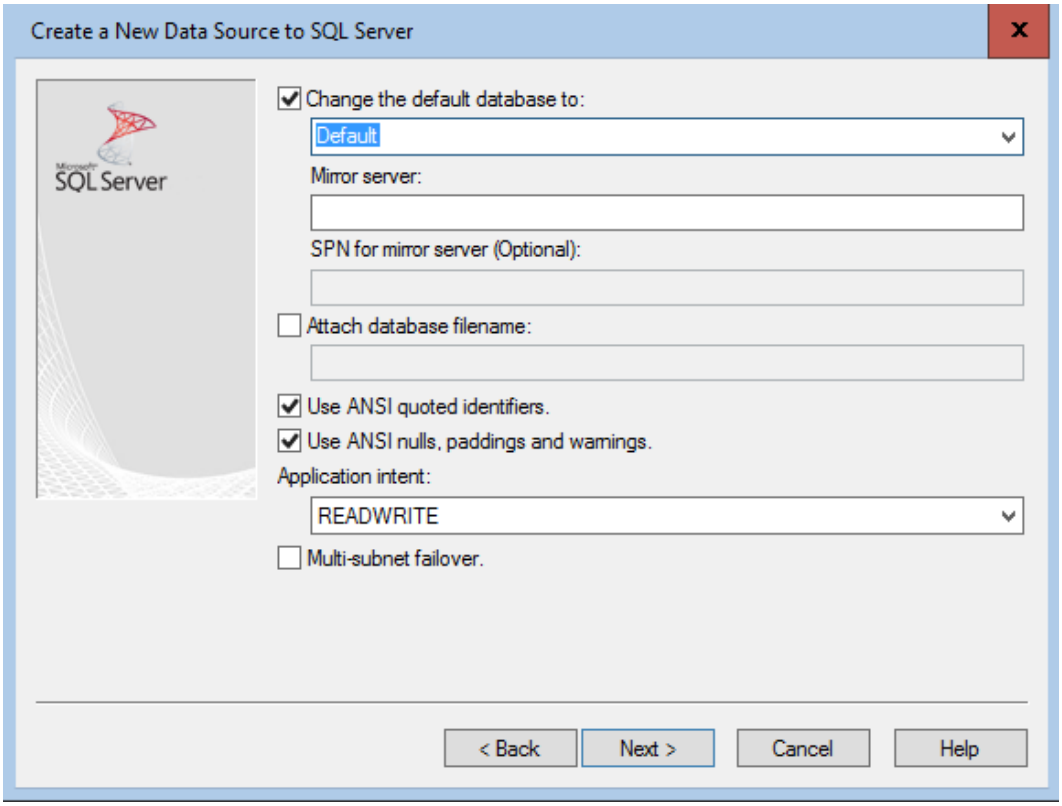
7. Select **ODBC Driver 13 for SQL Server** and click **Finish**.

   The Create a New Data Source to SQL Server box appears.

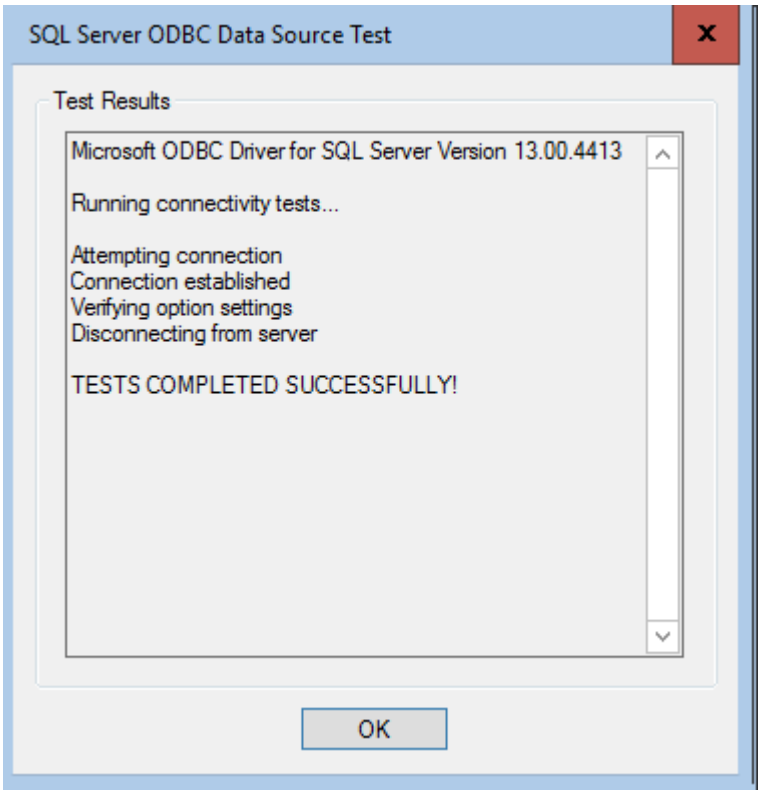


8. Specify the name, description, and the server you want to connect to.
9. Click **Next** and provide the user name and password for SQL Server Authentication.

10. Click **Next** and change the default database setting to the database you want to connect to.



11. Click **Next**.
12. Click **Finish** and then, click **Test Data Source**. Ensure that the test connection completes successfully.



13. Click **OK**.

Create a Microsoft Azure Synapse SQL ODBC connection

Create an ODBC connection to connect to Microsoft Azure Synapse SQL after you configure the Microsoft Azure Synapse SQL ODBC driver.

Perform the following steps to create a Microsoft Azure Synapse SQL ODBC connection on the **Connections** page:

1. In Administrator, click **Connections**.

   The Connections page appears.

2. Click **New Connection**.

   The New Connection page appears.

3. Configure the following connection details in the **Connection Details** section:

Property	Description
Connection Name	Name of the ODBC connection.
Description	Description of the connection.
Type	Type of the connection. Select the type of the connection as ODBC.

4. Configure the following connection details in the **Connection Properties** section:

Property	Description
Runtime Environment	The name of the runtime environment where you want to run the tasks.
User Name	User name of the Microsoft Azure Synapse SQL account.
Password	Password for the Microsoft Azure Synapse SQL account.
Data Source Name	Enter the name of the ODBC data source name that you created for the Microsoft Azure Synapse SQL database.
Schema	Microsoft Azure Synapse SQL schema name.
Code Page	Select the code page that the Secure Agent must use to read or write data.
ODBC Subtype	Enter the value of the ODBC Subtype field as Azure DW.
Driver Manager for Linux	The driver to which the Microsoft Azure Synapse SQL ODBC driver manager sends database calls. Select **unixODBC 2.3.4** to connect to Microsoft Azure Synapse SQL.

5. Click **OK**.

   You can use the configured connection in a mapping. You can then enable pushdown optimization on the Schedule tab in the mapping task.

Configuring pushdown optimization

Create a mapping that uses the configured ODBC connection. To optimize a mapping, add the mapping to a task, and then configure pushdown optimization in the mapping task

1. Create a mapping task.
2. In the **Pushdown Optimization** section on the Schedule tab, set the pushdown optimization value to **Full** or **To Source**.

When you run the mapping task, the transformation logic is pushed to the configured database.

To verify that the mapping was optimized, you can check the session log for the job. In Monitor, view the log for jobs.

Cross-schema pushdown optimization

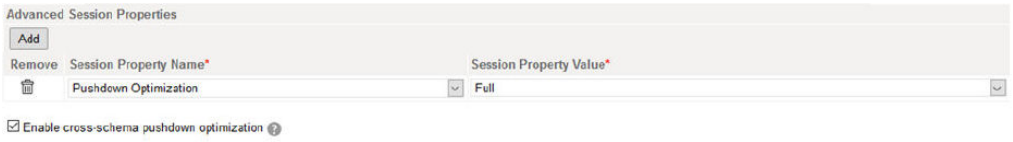
You can use cross-schema pushdown optimization for a mapping to read or write data based on different schemas within the same database.

To use cross-schema pushdown optimization, you must create two Microsoft Azure Synapse SQL ODBC connections and specify different schemas for each of the connections. Select the **Enable cross-schema pushdown optimization** check box in the advanced session properties of the mapping. By default, the check box is selected.

Consider the following steps to use cross-schema pushdown optimization:

1. Create an `Azure_DW1` ODBC connection and specify `DW_schema1` schema in the connection properties.
2. Create another `Azure_DW2` ODBC connection and specify `DW_schema2` schema in the connection properties.

3. Create a mapping and perform the following tasks:
   a. Select the `Azure_DW1` connection as source to read data from the source object using the `DW_schema1` schema.
   b. Select the `Azure_DW2` connection as target to write data to the target object using the `DW_schema2` schema.
   c. Select the **Enable cross-schema pushdown optimization** check box in the advanced session properties. The following image shows the **Enable cross-schema pushdown optimization** property:



   d. Run the mapping.

When you run the mapping, the Secure Agent reads data from the source object using `DW_schema1` and writes data to the target object using `DW_schema2`.

Note: Ensure that the schemas you specify for both the Microsoft Azure Synapse SQL ODBC connections belong to the same database.

Pushdown compatibility for Microsoft Azure Synapse SQL ODBC connection

You can configure the task to push transformations, variables, functions, and operators to the database.

When you use pushdown optimization, Data Integration converts the expression in the transformation by determining equivalent operators, variables, and functions in the database. If there is no equivalent operator, variable, and function, Data Integration processes the transformation logic.

Functions and operators

Microsoft Azure Synapse SQL supports full pushdown and source pushdown optimization.

The following table displays the functions that you can push to the Microsoft Azure Synapse SQL database by using source-side or full pushdown optimization:

Function	Function	Function
ABS()	LENGTH()	SQRT()
ADD_TO_DATE()	LOG()	STDDEV()
ASCII()	LOWER()	SUBSTR()
AVG()	LTRIM()	SUM()
CEIL()	MAKE_DATE_TIME()	SYSDATE()
CHR()	MAX()	SYSTIMESTAMP()
CONCAT()	MD5()	TAN()
COS()	MIN()	TANH()

Function	Function	Function
COUNT()	MOD()	TO_BIGINT
DATE_DIFF()	POWER()	TO_CHAR(DATE)
DECODE()	REPLACECHR()	TO_CHAR(NUMBER)
EXP()	REPLACESTR()	TO_DATE()
FIRST()	ROUND(NUMBER)	TO_DECIMAL()
FLOOR()	RTRIM()	TO_FLOAT()
GET_DATE_PART()	SET_DATE_PART()	TO_INTEGER()
IIF()	SIGN()	TRUNC(NUMBER)
INSTR()	SIN()	UPPER()
ISNULL()	SINH()	VARIANCE()
LAST()	SOUNDEX()	

You can use the following operators in full pushdown and source pushdown optimization:

```
+ - * / % = > < >= <= <> != ^= NOT AND OR
```

Function Restrictions

- The Microsoft Azure Synapse SQL aggregate functions accept only one argument, which is a field set for the aggregate function. The agent ignores any filter condition defined in the argument. In addition, ensure that all fields mapped to the target are listed in the `GROUP BY` clause.
- To push the `TO_CHAR()` function to the Microsoft Azure Synapse SQL database, you must define the date and format arguments.
- When you push the `SYSTIMESTAMP()` and `SYSDATE()` functions to the Microsoft Azure Synapse SQL database, do not specify any format.
  The Microsoft Azure Synapse SQL database returns the complete timestamp. `SYSDATE` works only without the brackets `()`. If used, shows as invalid expression.
- You cannot push the `TO_BIGINT()` or `TO_INTEGER()` function with more than one argument to the Microsoft Azure Synapse SQL database.
- When you push the `REPLACECHR()` or `REPLACESTR()` function to the Microsoft Azure Synapse SQL database, the agent ignores the caseFlag argument.
  For example, both `REPLACECHR(false, in_F_CHAR, 'a', 'b')` and `REPLACECHR(true, in_F_CHAR, 'a', 'b')` return the same value.
- To push `INSTR()` to Microsoft Azure Synapse SQL database, you must only define string, search_value, and start arguments. Do not use occurrence and comparison_type arguments.
- Use the following date formats for the `TO_DATE()` function:

  - `YYYY-MM-DD HH24:MI:SS.NS`
  - `YYYY-MM-DD HH12:MI:SS.NSAM`
  - `MON DD YYYY HH12:MI:SS.NSAM`

- MON DD YYYY HH24:MI:SS.NS
- DD MON YYYY HH12:MI:SS.NSAM
- DD MON YYYY HH24:MI:SS.NS
- MM/DD/YY HH12:MI:SS.NSAM
- MM/DD/YY HH24:MI:SS.NS
- MM/DD/YYYY HH12:MI:SS.NSAM
- MM/DD/YYYY HH24:MI:SS.NS
- HH24:MI:SS.NS
- HH12:MI:SS.NSAM

- To push the `SET_DATE_PART()` function to the Microsoft Azure Synapse SQL database, use the following date data types as arguments:
  - datetime
  - datetimeoffset
  - datetime2
  - smalldatetime

  Use the following formats for date data types:
  - YYYY, YY, YY, Y
  - MM, MON, MONTH
  - D, DD, DD, DY, DAY
  - HH, HH12, HH24
  - MI
  - MS
  - SS

  Note: `NS` and `US` formats are not applicable to `SET_DATE_PART()`.
- To push the `ADD_TO_DATE()` function to the Microsoft Azure Synapse SQL database, use the following date data types as arguments:
  - date
  - datetime
  - datetimeoffset
  - datetime2
  - smalldatetime
  - time

  You can use the following formats for date data types:
  - YYYY, YY, YY, Y
  - MM, MON, MONTH
  - D, DD, DD, DY, DAY
  - HH, HH12, HH24
  - MI
  - MS

- SS
- NS: applicable to datetimeoffset, datetime2, and time
- US

- To push the MAKE_DATE_TIME() function to the Microsoft Azure Synapse SQL database, use the following date data types as arguments:
  - date
  - datetime
  - datetimeoffset
  - datetime2
  - smalldatetime
  - time

  You can use year, month, day, hour, minute, second, and nanosecond with appropriate return date types.

Transformations with Microsoft Azure Synapse SQL ODBC connection

The following table lists the transformations that you can push using source of full pushdown:

Transformations	Pushdown
Aggregator	Source, Full
Expression	Source, Full
Filter	Source, Full
Joiner	Source, Full
Lookup	Source, Full
Router	Full
Union	Source, Full

Source transformation

Mappings enabled for pushdown optimization has the following restrictions for sources:

- When you read data that contains reserved keywords from Microsoft Azure Synapse SQL, ensure that you add the keywords to the reswords.txt file in the agent machine. The reswords.txt file is available in the following path:
  <Secure Agent installation directory>\downloads\package-ICSAgent_RXX.X\package\ICS\main\bin\rdtm
- You cannot use the ORDER BY clause in a source custom query unless you also specify a TOP, OFFSET, or FOR XML clause in the query.
- The datetimeoffset datatype is applicable only for passthrough mappings.

Target transformation

You can use an existing Microsoft Azure Synapse SQL target or create a Microsoft Azure Synapse SQL target to write data.

Mappings enabled for pushdown optimization has the following restrictions for targets:

- You cannot use the upsert operation when you write data to Microsoft Azure Synapse SQL.
- The update override target property is not applicable in a target operation.
- If you run a mapping to create a Microsoft Azure Synapse SQL target at runtime and drop the target that was created on Microsoft Azure Synapse SQL using the drop table command, and then rerun the same mapping, the Secure Agent instead of creating the target and then writing data to that target, does not trigger the create query and results in an error.

Author

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